

B.Tech. Degree VI Semester Regular/Supplementary Examination May 2024

ME 19-205-0607 – ENERGY CONSERVATION AND ENVIRONMENT PROTECTION

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the overview of World Energy Scenario and energy economics.
CO2: Study the Importance of energy management and energy auditing.
CO3: Learn about the renewable energy sources.
CO4: Understand the Environmental Impacts of energy use.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
1. (a) Classify Energy sources.	3	L1	1	1,3,7,10	
(b) Define simple payback period.	3	L2	1	1,3,7,10	
(c) Explain the need for energy audit.	3	L1	2	1,6,7,8,9	
(d) Write any three energy conservation opportunity in pumping system.	3	L1	2	1,6,7,8,9	
(e) Write the criteria for selection of wind mill installation.	3	L1	3	1,3,6,7, 11,12	
(f) Explain principles of cogeneration.	3	L1	3	1,3,6,7, 11,12	
(g) Define the term Environmental Auditing.	3	L1	4	1,7,8,10, 11	
(h) What are the major Environmental effects of building Hydro Power Plants?	3	L1	4	1,7,8,10, 11	

PART B

(4 × 12 = 48)

II.	Explain about Global Primary energy reserves.	12	L2	1	1,3,7,10
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OR

(P.T.O.)

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Marks BL CO PO
12 L3 1 1,3,7,10

✓ III

Using NPV analysis technique, evaluate the financial merits of the two proposed projects given. Annual discount rate is 8% for both the projects.

	Project 1	Project 2
Capital cost (₹)	30000	30000
Year	Net annual savings (₹)	Net annual savings (₹)
1	+6000	+6600
2	+6000	+6600
3	+6000	+6300
4	+6000	+6300
5	+6000	+6000
6	+6000	+6000
7	+6000	+5700
8	+6000	+5700
9	+6000	+5400
10	+6000	+5400
Total net savings at end of 10 th year	+60000	+60000

✓ IV

Explain the steps and methodology for conducting the detailed Energy Audit.

12 L2 2 1,6,7,8,9

OR

- V. (a) List energy efficiency opportunities for refrigeration and air conditioning systems.
(b) List various heat recovery devices and explain the working of Heat Wheels with diagram.

6 L2 2 1,6,7,8,9

6 L2 2 1,6,7,8,9

✓ VI

- (a) With a diagram explain IC engine co-generation system.

6 L2 3 1,3,6,7,11,12

- (b) Explain pinch, target setting and composite curves of heat exchangers.

6 L2 3 1,3,6,7,11,12

OR

✓ VII

Explain working of various types of Geothermal power plants with diagrams.

12 L2 3 1,3,6,7,11,12

✓ VIII

- (a) Explain the necessity of testing engine of vehicles and write the procedures for the test.
(b) How do you describe the term emission factor and its significance with the help of an example?

6 L2 4 1,7,8,10,11

6 L2 4 1,7,8,10,11

OR

IX

How the energy usage deteriorates the environment? Explain with air and water pollution.

12 L2 4 1,7,8,10,11

Blooms's Taxonomy Level's

L1 - 17.5%, L2 - 72.5%, L3 - 10%.
